MILITARY SPECIFICATION

MECHANISM, RECOIL, 155MM HOWITZER: M45

This specification is approved for use by U.S. by Armament Research and Development Command (Rock Island Arsenal), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hydro-pnematic recoil mechanism designed for use with the 155 millimetre (mm) howitzer, M198,

2. APPLICABLE DOCUMENTS

2.1 Issues of' documents. The following documents, of the iolowing documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specifed herein.

SPECIFICATIONS .

Federal	
PPP-T-42	Tape, Packaging/Masking, paPer
PPP-T-60	Tape: Packaging, Waterproof
cCC-C-429	Cloth, Osnaburg, Cotton
Military	
MIL-P-116	Preservation-Packaging, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-130	Paper, wrapping, Laminated and Creped
MIL-C-5501	Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Resaearch and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self addressed Standardization Document Improvment propoasl appearing at the end of this document or by letter.

FSC 1025

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MIL-H-5606	Hydraulic Fluid, Petrolem, Base; Aircraft, Missiles and Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base, for Preservation and Testing
MIL-H-6088	Heat Treatment of Aluminum Alloys
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice, Process for)
MIL-S-7742	Screw Threads, Standard, optimum Selected Series: General Specification for
MIL-R-11468	Radiographic Inspections, Soundness Requirements for Arc and Gas Welds In Steel
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-E-16053	Electrodes, welding, Bare, Aluminum Alloys
MIL-C-16555	Coating Compound, Strippable, Sprayable
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-W-45205	Welding, Gas Metal - Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-R-45774	Radiographic Inspection, Soundness Requirements for Fusion Welds in Aluminum and Magnesium Missile Components
MIL-Q-45970	Quality Assuance for Weapons and Support Materiel

STANDARDS

Federal

Fed. Std. No. 595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-sTD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property

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MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-248 - Welding and Brazing Procedure and performance Qualification
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1186 - Cushioning, Anchoring Bracing, Blocking and Waterproofing: With Appropriate Test Methods
MIL-STD-1252 - Inertia Friction Welding Process, Procedure and Performance Qualification
MIL-sTD-1261 - Welding Procedures for Construction Steels

DRAWINGS

Rock Island Arsenal
F12007800 - Recoil Mechanism, 155MM Howitzer, M45

Product drawings referenced in this specifica
F12007800.)

PUBLICATIONS

Rock Island Arsenal
LQ12007800 - Quality Requirements List, Recoil Mechanism, 155MM Howitzer) ,45
EL12007800-I - List of Inspection Equipment Numbers, Recoil Mechanism, 1551MM Howitzer, M45

U.S. Army
AR702• 3 Army Materiel Reliability, Availability and Maintainability (RAM)

(Copies *of* specifications, standards>drawings and publications required by contractors in connection with specific procumbent functions should. be obtained from the procuring activitY or as directed by the cOntracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extend specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids on request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI B46,1 - Surface Texture
ANSI Y1.4.5 - Dimensioning and Tolerancing

(Applications for copies should be addressed to The American Society of Mechanical Engineers, Limited Engineerin Center, 345 East 47th Street, New York, NY 10017.)

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AMERICAN WELDING SOCIETY (AWS)

AWS AZ, 4 - Symbols for Welding and
Nondestructive Testing

(Application for copies should be addressed to The American Welding Society,2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread
Standards For Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The recoil mechanism shall conform to the materials, construction and design requirements specified herein and on Drawing F12007800.

3.2.1. Heat treatment. Unless otherwise specified on the applocable drawings, heat treatment for aluminum alloys shall conform to MIL-H-6088 and for steel shall conform to MIL-H-6875.

3.2.2 Dimensions. Dimenisions and tolerances of parts and assemblies of the recoil mechanism shall conform to the detailed drawings of Drawing F12007800, Interpretation of the dimensions and tolerances shall be in accordance with ANSI Y14.5.

3.2.3 Surface texture. Surface texture shall be as specified on the applicable drawings. Interpretation of surface roughness symbols shall be as defined in ANSI B46.I.

3.2.4 Protective finishes.

3.2.4.1 Inorganic finishes. Parts requirng an inorganic finish shall be finished or coated as indicated on the applicable drawings and in accordance with MIL-STD-171.

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3.2. 4.2 Painting. The parts and assemblies shall be painted as specified on the drawing and. in accordance with MIL-STD-171 . The visible surfaces of attaching hardware (nuts, screws, washers, etc.), exposed unpainted surfaces of mounting faces; or blemished areas shall be primed and painted to match surounding surfaces after final assembly. To avoid necessity of painting exposed surfaces of cadmium plated hardware (including screws, nuts and washers), the hardware may be dyed with a supplementary chromate finish, Olive Drab 24087 of Fed. Std. No. 595.

3.2,5 Welding, Interpretation of weld symbols specified on the applicable drawings shall be as defined in AWS A2.4. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedures and test specimens of' the applicable welding specifications, Qualification of are welding procedures and welders shall be in accordance with MIL-STD-248. Qualification of inertia friction welding shall be in accordance with MIL-STD-1252. All plug welds shall be flush and free of snags or protrUSiOnS. Grinding flush is not required unless specified on the drawing or in the welding specification. Weld splatter and similare excess weld shall be removed.

3.2.5.1 Arc welding, aluminum alloy. Welding of aluminum alloy parts shall conform to MIL-W-45205. Type 5356 welding electrod of MIL-E-16053 shall be used for 5000 series aluminum alloy welds. Type 5183 or 5556 welding electrodes of MIL-E-16053 may be used if me 5356 welding electrode is unavailable. When welding aluminum alloy parts other than 5000 series aluminum alloy, the welding shall be in accordance with the specification referenced on the applicable drawings.

3.2.5.2 Arc welding, ferrous metals. Welding of steel parts and assemblies shall be in accordance with weld classification of MIL-STD-1261 referenced on the applicable drawings.

3.2.5.3 welding, inertia. Inertia friction welding shall be in accordance with MIL-STD-1252.

3.2.5.4 Magnetic particle inspection. When magnetic particle inspection is indicated on the drawing, it shall be in accordance with MIL-I-6868.

3.2.5.5 Penetrant inspection. When penetrant inspection is indicated on the drawing, it shall be in accordance with MIL-I-6866.

3.2.5.6 Radiographic inspection. Radiographic inspection shall be in accordance with MIL-STD-453. Aluminum inspection shall be to standards of MIL-R-45774 and steel inspection to standards of MIL-R-11468 as referenced on the applicable drawings.

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3.2.6 Threads. Threads shall conform to the thread series and class indicaed on the applicable drawing, and be in accordance with Handbook H28. Unless otherwise specified, coated or plated Class 3 threads shall conform to the coated thread data specified in MIL-S-7742.

3.2.6.1 Torque, threaded fasteners. Unless otherwise specified on the applicable drawing, threaded fasteners shall be torqued in ccordance with the following torque requirements:

Thread Size	Torque	
.112 - 48UNF	9-1o	inch-pounds
.138 - 321unc	16-18	inch-pounds
.164 - 32UNC	30-33	inch-pounds
.190 - 24UNC	40-45	inch-pounds
l 190 - 32UNF	45-50	inch-pounds
,250 - 20UNC	8-10	foot-pounds
.375 - 16UNC	30-33	foot-pounds
l 500- 13UNC	75-80	foot-pounds
.625 - llUNC	150-165	foot-pounds

3.3 performance characteristics.

3.3.1 General. The recoil mechanism shall be filled with hydraulic fluid, bled. and filled until all air and measurable moisture are eliminated. The initial and subsequent filling of the hydraulic system shall be with hydraulic oil conforming to MIL-H-5606 or MIL-H-6083 and filtered through a 5 micron filter. The completed recoil mechanism shall conform to the characteristics specified on Drawing F12007800 and the requirements of 3.3.2 through 3,3.7. Testing shall be as specified in 4.5.3.2.

3.3.2 Function firing. The recoil mechanism shall withstand function firing as specified in Table I without evidence of malfunction or failure. The sewuence shall be maintained as listed in Table I to assure check of variable recoil mechanism transition from short to long recoil. The counter recoil terminal velocity shall not exceed an average of 6.oo inches/second for the last 6.oo inches of travel. After function firing, the recoil mechanism shall meet the requirements specified in 3.3.3 through 3.3.7.

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TABLE I

FUNCTION FIRING 1/

No. of Rounds	Prop Charge	Zone	Proj. Inert	Quadrant Elevation and/or Variable Recoil Setting	Recoil Length (Inches)	Cycle Time (Seconds)
1	M4A2	7	2/	Less than 300 mils-Long Recoil	40-44	2 to 4
1	M203	8	2/	Less than 300 mils-Long Recoil	64-68	2 to 4
1	M203	8	2/	Greater than 800 mils-Short Recoil	50-54	2 to 4
1	M203	8	2/	Less than 300 mils-Long Recoil	64-68	2 to 4

NOTES:

1/ All charges shall be conditioned and fired at 135° F to 145° F.
2/ Any standard projectile loaded to 103± 1 pounds.

3.3.3 Friction. Friction oof the recoil rod, when assembled in the recoil cylinder and the floating piston assembly, shall be within the limits specified in Table 11 after assembly, and Table III after function

TABLE II
(After Assembly)

FRICTION IN POUNDS PER SQUARE INCH(PSI)

HYDRAULIC PRESSURE

Part Name	Max. Limits
REPLENISHER, PISTON D12007613	35 PSI
ROD, RECOIL F12007661	150 PSI
RECUPERATOR, FLOATING PISTON ASSY.	60 PSI

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TABLE III
(After Function Firing)

FRICTION IN POUNDS PER SQUARE INCH(PSI)

HYDRAULIC PRESSURE

Part Name	Max. Limits
REPLENISHER, PISTON D12007613	50 PSI
ROD' RECOIL F12007661	220 PSI
RECUPERATOR, FLOATING PISTON ASSY.	80 PSI

3.3.4 Nitrogen leakage. when the system is pressurized, the nitrogen filling end of the recuperator cylinder, the breech end of the air cylinder and the connections for the cross-over tube between the recuperator and air cylinder shall show no evidence of nitrogen leakage when observed for five minutes.

3.3.5 Reserve oil indicator. The reserve oil indicatior piston stem shall function without evidence of binding or sticking-throughout its entire range of travel. There shall be no evidence of oil leakage other than oil wetting around the piston stem.

3.3.6 Foamed oil. and contaminated oil.

3.3.6.1 Foamed oil. Ater function firing, oil withdrawn slowly from the recoil mechanism into a straight sided glass beaker shall be in accordance with the following: Within the first minute afer withdrawal, the oil shall be five percent or less foam by volume; when exposed to the atmosphere for 15 minutes, the aeration bubbles shall disappear.

3.3.6.2 Contaminated oil. After fiction firing, a minimum of two ounces of oil shall be removed from the purge hole in the recuperator cylinder into a straight sided glass beaker. Metallic and non-metallic particles allowable in the oil sample shall not exceed 400 microns in size. Insides of recoil cylinders shall show no evidence of scoring after function firing.

3.3.7 Housing and regulator assembly. The torque requiredto rotate the housing and regulator assembly, Drawing F12007872, between the positive stops shall not exceed 20 foot-pounds after function firing.

3. 3.4 Reliability, maintainability and durability. Recoil maechanisms manufactured in accordance with this specification shall be free from any defects that compromise, limit or reduce the end item howitzer or component capability specified herein and shall be subject to examination and

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test by the Government to prove howitzer capabilities and performance in accordance with the following:

3.4.1 Reliability. The reliability requirement is a specified value of 1100 mean rounds between failure and a minimum acceptable . value of 700 mean rounds between failure as defined in AR 702.3.

3.4.2 Maintainability. The maintainability requrement is 0.5 hours for mean time to repair (MTTR) at organizational and 2.0 hours for direct support.

3.4.3 Durability. The durability requrement is 15,000 Equivalent Fu1l Charge (EFC) rounds without rebuild or overhaul.

3.5 Identification and instructional marking. Identification and instructional markings shall be applied in accordance with Drawing F12007800. Other marking, including application of part numbers and instructional marking, shall be as specified on the applicable drawings and in accordance with MIL-STD-130. Serial numbers shall be furnished by the procuring activity (see 6.1).

3.6 Workmanship. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons, Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edes, nicks, scratches, burrs, deformations and missing operations which may affect serviceability, functioning, operation, appearance, or safety Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval Of the contracting offiCer.

4, QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in. the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements. specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such. inspection.s are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1,1 The contractor shall perform, as a minimum, inspections in accordance with the Specifications, qualitY assuance Provisions> MIL-Q.-4597O and such documents as specified in LQ12007800, and the contract. These minimum inspections shall not be construed as relieving

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the contractor of his responsibilities under terms of the contract to furnish the Government with items complying with and conforming to the requirements of the contract, drawings and specifications.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requrements specified herein are as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as necessary to determine that all requirements of the contract have been complied with. One of the first article items will be subjected to a reliability, maintainability, and durability demonstration (see 4,7), performed by the Government for informational purposes.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size, and presentation of inspection lots shall be in accordance with MIL-STD-105. Recoil mechanisms shall be assembled from lots of component parts that have complied with all inspection requirements specified herein.

4.5.2 Examination.

4.5.2.1 Recoil mechanisms. Visually and manually examine each recoil mechanism to determine conformance with the applicable requirements. Each step in the examination shall include a visual examintions for proper cleaning, the presence of the specified protective coating, and to determine the General quality, completeness of manufacture, assembly, and workmanship (see 3.6) Visually examine all marking to assure that they are correct and legible (see 3.5).

4.5.2.2 Parts and assemblies. Parts and assemblies shall be visually and manually examined to determine compliance with the materials, construction, and design requirements of 3.2 and the applicable requirements of 3.2.1 through 3.2.6. Visually examine for marking (see 3.5) and workmanship (see 3,6). Parts and assemblies failing to comply with the requirements shall be rejected.

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4.5.2.2.l Heat treatment. Examination and tests of heat treated components shall be in accordance with MIL-H-6088 or MIL-H-6875, as applicable.

4.5.2.2.2 Surface texture. When surface texture is is specified on the drawing, examination and interpretation shall be in accordance with ANSI B46.1.

4.5.2.2.3 Welds and welding. Examination and tests of welded parts and welding procedures for aluminum alloys shall be in accordance withMIL-W-45205. Examination and testof welding procedures for steel parts shall be in accordance with MIL-STD-243 or as otherwise specified. on the applicable drawings.

4.5.2.2.4 Magnetic particle inspection. When magnetic particle inspection is specified on the drawing. the inspection shall be in accordance with MIL-I-6868.

4.5.2.2.5 Penetrant inspection. When penetrant inspection is specified on the drawing, the inspection shall be in accordance with MIL-I-6366.

4.5.2.2.6 Radiographic inspection. When radiographic inspection is specified on the drawing, the inspection shall be in accordance with the specifications referenced on the drawing.

4.5.2.2.7 Protective finishes. Examination of painted surfaces and inorganic finishes shall be in accordance with the specifications specified on the applicable drawings and MIL-STD-171.

4.5.2.2.8 Marking. Visually examine the identification plate for compliance with the applicable drawing, for secure assembly, serialization and to assure that all pertinent data has been applied.

4.5.3 Testing.

4.5.3,- Failure data. Under otherwise specified herein, all tests shall be conducted on a complete recil mechanism, If test requirements cited herein are not complied with, acceptance of the recoil mechanism shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical, and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect (s) in future production.

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c. Examine recoil mechanisms, partially assembled recoil-mechanisms, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned recoil mechanisms for retest.

4.5.3.2 Performance characteristics testing. Each completely assembled recoil mechanism shall be tested for performance characteristics (see 3.3 and 3.3.1 thru 3.3.7) using the test methods specified in 4.6. Failure of the recoil mechanism to comply with the requirements shall be cause for rejection.

4.5.4 Inspection equipment. Unless otherwise specified herein, responsibility for acquisition, calibration, maintenance anddisposition of acceptance inspection and test equipment prescribed on List of Inspection Equipment Numbers EL12007800-I and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.5 Inspection of packaging. Unless otherwise specified (see 6.1), inspection to determine compliance with cleaning, preservation, packing and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-P-14232.

4.6 Test methods. Each recoil mechanism shall be checked for nitrogen pressure, recoil oil, and other operating paramenters specified on Drawing F12007800 prior to subjecting the recoil. mechanism to the function firing test.

4.6.1 Function firing test. Each recoil mechanism shall be function fired as specified in 3.3.2. If simulation equipment is used, all testing shall be done on a test stand at zero degrees with the variable recoil locked for long recoil or short recoil, as specified. Simulation equipment shall be capable of providing an impulse of 10,800 pound-seconds applied at 8 ± 3 milliseconds. A typical recoil oil pressure-time trace is shown in Figure 1. Upon completion of function firing, the recoil mechanism shall be subjected to-the tests specified in 4.6.2 thru 4.6.7.

4.6.2 Friction test, Pressure readings shal be taken and recorded on the replenisher, the recoil rod assembly and the recuperator floating piston assembly to determine that friction of these components does not exceed the maximum limits specified in 3.3.3.

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4.6.2.1 Replenisher friction. The replenisher friction shall be measured by applying pressure through the oil filling valve and measuring the maximum pressure.

4.6.2.2 Recoil cylinder friction. The oil line between the Recuperator and the replenisher shall be blocked and one of the recoil rods shall be released and extended 12 inches while leaving the other fully retracted. The recoil cylinder friction shall then be measured by applying pressure through the oil filling valve and measuring the maximum pressure necessary to fully retract the extended recoil rod. The other recoil cylinder shall then be tested using the same procedure.

4.6.2.3 Recuperator friction. The oil line between the recuperator and replenisher shall be blocked and, both recoil rods shall be fully retracted. The recuperator friction shall then be measured by applying pressure through the oil filling valve and measuring the maximum pressure necessary topump a minimum of 95 ounces of oil into the system. The system shall then be filled back and the maximum pressure measured. One recuperator friction is one-half the difference between the two measured pressures.

4.6.3 Nitrogen leakage test. A solution of soap and water shall be applied to the nitrogen filling end of the recuperator cylinder, the breech end of the air cylinder and the connections for the cross-over tube between the recuperator and air cylinders when the System is pressurized. There shall be no evidence of soap bubbles when observed for five minutes.

4.6.4 Reserve oil indicator test. Prior to replacement Of the reserve oil, the reserve oil indicator shall be checked for freedom of movement throughout its entire range of travel. Upon replensihing the reserve oils the reserve oil indicator shall be visually checked for oil leakage around the piston stem.

4.6.5 Foamed oil test. Within five minutes after funcition firing, a minimum of four ounces of oil shall be withdrawn slowly from the recoil mechanism into a glass beaker. The recoil mechanism shall be rejected, if, after exposure to the atmosphere for one minue, the quantity of foam exceeds five percent by volume, and when aeration bubbles in the oil do not disappear after exposure to the atmosphere for 15 minutes.

4.6.6 Contaminated oil test. Within five minutes after function firing, two ounces of oil shall be drained from the purge hole of the recuperator cylinder. Particles contained in the sample shall be measured to determine compliance with the maximum limits specified in 3.3.6.2. After function firing, bleed oil until recoil rods are fully extended and. remove recoil cylinder breather end caps. Remove excess oil from recoil cylinder and visually examine for soring using a light source.

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4.6.7 Housing and regulator assembly test. After firncion firing. the operating linkage shall be disconnected from the housing and regulator assembly and a measurement taken of the torque required to rotate the assembly between the positive stops.

4.7 Reliability, maintainability and durability demonstration The howitzer demonstration will be conducted in accordance with the operational mode summary and reliability failure definition in the M198 scoring criteria. The test will be conducted by the Government for informational purposes and shall provide the equivalent of a 2,500 EFC round reliability test and a 15,000 EFC round durability test. The two tests maybe cabined. Maintenance data will be collected according to the scoring criteria, and the MTTR will be calculated according to the scoring criteria. howitzers exhibiting a reliagil, maintainability or durability failure shall be cause for the contractor to perform a failure analysis in accordance with 4.5.3.1. The Government representative shall verify the results of the failure analysis and assure deficiencies causing the failure have been corrected on all weapons and components in process at the contractor's plant.

5. PACKAGING

5.1 Pilot pack. The pilot pack shall consist of a complet coil mechanism preserved in accordance with the requirement specified herein for the level of protection specified (see 6.1), packed Level B and forwarded as specified in 3.1.

5.2 preservation. Preservation shall be Level A or B, as specified (see 6.1). Materials methods, processes and, procedures specified herein shall conform to the requirements of MIL-P-116 and MIL-P-14232.

5.2.1 Level A.

5.2.1.1 Cleaning. Unless otherwise specified herein, all unpainted ferrous surfaces shall be cleaned by process C-3. Painted and non-ferrous surfaces shall be cleaned by process C-1. Hydraulic system assemblies and components shall be cleaned in accordance with applicable drawing requirements.

5.2.1.2 Drying. Unless otherwise specified herein, all cleaned surfaces shall be dried by any applicable procedure of MIL-P-116 which is non-injurious to the item.

5.2.1.3 Preservative application, oils or lubricants *removed or* contaminated during cleaning shall be reapplied in accordance with applicable drawing requirements. Unpreserved exteriof merre surfaces shall be preserved with grease, Type P-n of MIL-P-116. All interior access opening of hydraulic system assemblies and components

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shall be sealed with. caps or plugs, as applicable, in accordance with MIL-C-5501.

5.2.1.4 Unit pack. The unit pack for mechanism shall be Method I and as specified herein:

a. Wrap the entire recoil mechanism with barrier material conforming to Type I, Grade A, Class 2 of MIL-B-121. Secure wrap to adjacent surfaces with tape conforming to PPP-T-42.
b Overwrap barrier material with osnaburg cloth conforming -to Class 2 of CCC-C-429. Secure the overwrap with tape conforming to Type 111, Class 2 of PPP-T-60.
c. Overspray the osnaburg cloth at least one inch onto adjacent surfaces with strippable compound conforming to Type II, Class 1 of MIL,-C-16555, applied in successive layers until a thickness of 0.035 ± .005 inch is obtained.
d. Provide a one inch opening on the underside of the sprayed recoil mechanism to allow for drainage of entraped moisture.

5.2.2 Level B.

5.2.2.1 Cleaning, drying and preservative application. Cleaning drying and preservative application shall be the same as specified for Level A (see 5.2.1.1, 5.2.1.2 and 5.2.1.3)

5.2.2.2 Unit pack. The unit pack for each recoil mechanism shall be the same as specified for Level A (see 5.2.1.4) with the following exception : Delete the overwrap of osnaburg cloth (CCC-C-429) and strippable compound (MIL-C-16555) and substitute an overwrap of Type 1 of MIL-F-130, secured to adjacent surfaces with tape conforming to PPP-T-60.

5-3 Packing, packing shall be Level Aor B, as specified (see 6.1).

5.3.1. Level A. Each-unit-pack. shall be located in minimum size wood shipping container conforming to Type II, Style A, Class 2 of MIL-B-26195 with. plywood superstructure. Blocking and bracing of the container contents shall be in accordance with MIL-STD-1186. Closure of the container shall be in accourdance with MIL-B-26195.

5.3.2 Level B. Packing shall be the same as specified for Level A (see 5.3.1)except the wood shipping container shall conform to Type I of MIL-B-26195 in lieu of Type II.

5.4 Marking.

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5.4.1 Levels A and B. The shipping container and contents shall be marked in accordance with MIL-STD-129. The words "THIS SHIPPING CONTAINER HAS REUSE VALUE" shall be stenciled on the inside of the cover and at the upper third of one side and one end of each shipping container.

5.4.2 Special Marking. The followingg warning shall be stenciled on the ends and side panels of the assembled shipping container: "CAUTION: LIFT BY BASE ONLY"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

Title, number and date of this specification.
a: List of drawings, specifications and publications pertinent to the recoil mechanism, showing applicable revision dates.
c. Shipping instructions for first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Block of serial numbers (see 3.5).
e. Packaging inspection, if different (see 4.5.5).
f. Selection of applicable level of protection, A or B (see 5.1, 5.2 and 5.3).

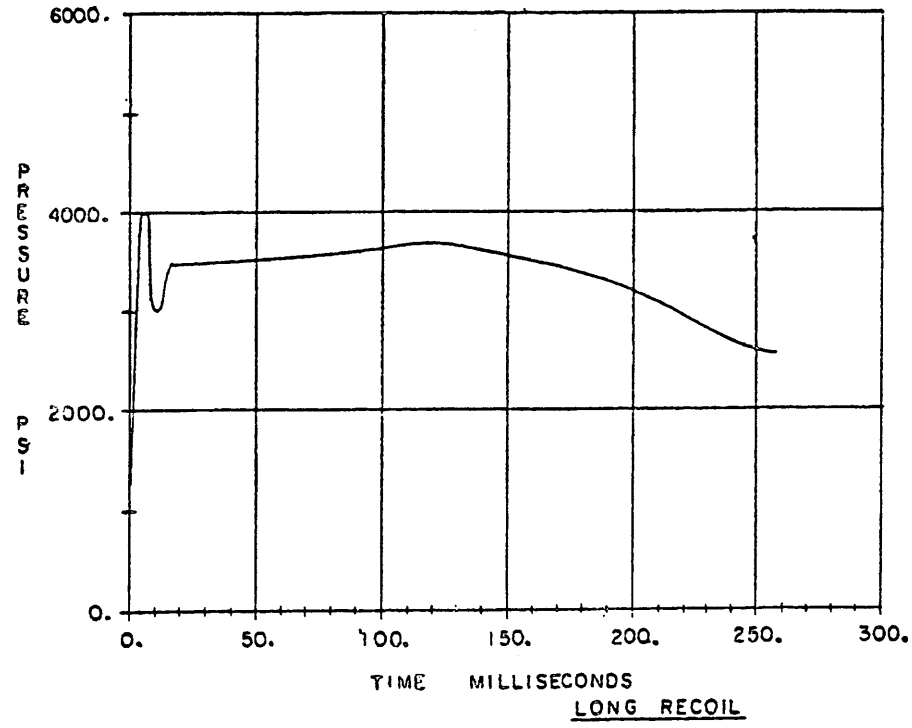
6.2 Assembly area cleanliness. Recoil mechanism assembly areas should be air conditioned using a unit with a 50 micron filter. The area temperature should be maintained between 68° to 72°F. The air flow rate per cubic inch per minute should maintain area air pressure between .75 to 1.5 inches of water. Relative humidity should not exceed 40-50 percent. Areas should be clean and free of dust, lint and abrasive materials that may become entrapped in assembled mechanisms. No grinding, filing, scraping or any operation producing abrasive dust should be performed in the assembly area.

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Project No.
1025-A118

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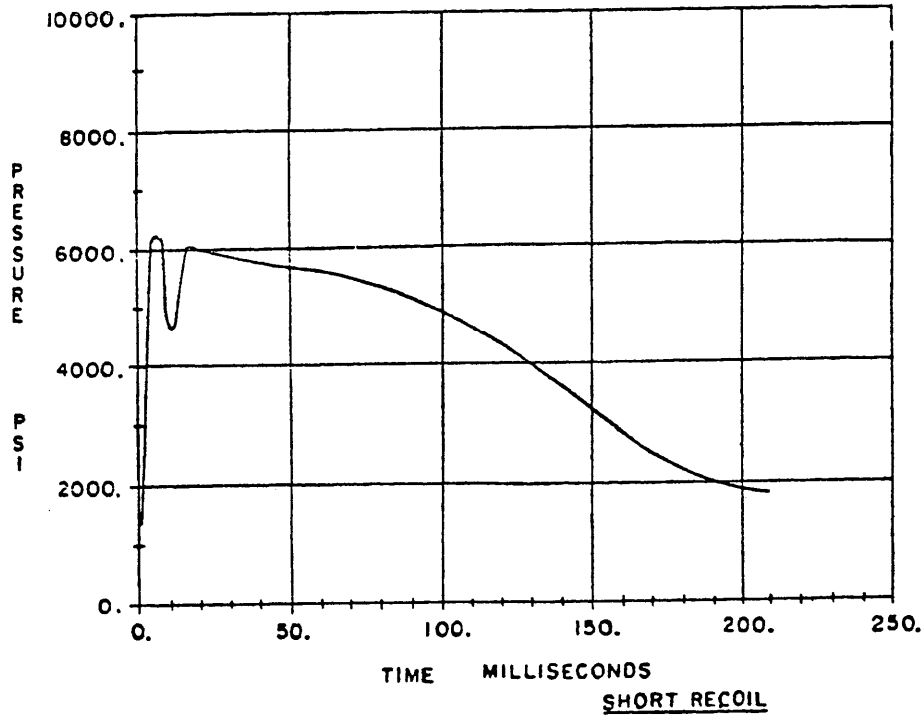


FIGURE 1: RECOIL OIL PRESSURE—TIME TRACE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-M-45986(WC); Mechanism, Recoil, 155MM Howitzer: M45

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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MILITARY SPECIFICATION

MECHANISM, RECOIL, 155MM HOWITZER: M45

This amendment forms a part of Military Specification MIL-M-45986 (WC), dated 15 February 1977, and is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 6

3.2.6.1 Torque, threaded fasteners: Delete the table and insert the following new table:

"Thread Size	Torque
No. 4-48UNF	6-8 in-lb
No. 6-32UNC	7-9 in-lb
No. 8-32UNC	14-17 in-lb
No. 10-24UNC	20-25 in-lb
No. 10-32UNF	20-25 in-lb
.250-20UNC	70-80 in-lb
.375-16UNC	20-24 ft-lb
.500-13UNC	50-60 ft-lb
.625-llUNC	90-119 ft-lb"

3.3.2 Function firing: Delete in its entirety Delete in its entirety and substitute:

"3.3.2 Function firing. The recoil mechanism shall withstand the function firing test (powder (powder gymnastication) as specifies in 4.6.1 without evidence of malfunction or failure and shall meet the criteria of 3.3.2.1 through 3.3.2.5. After function firing, the recoil mechanism shall meet requirements specified in 3.3.3 through 3.3.7.

FSC 1025

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3.3.2.1 Long recoil length Maximum Long recoil length shall not exceed 71 inches.

3.3.2,2 Short recoil length Maximum short recoil length shall not exceed 50 inches

3.3.2.3 Cycle time. Recoil cycle time shall not be less shall not be loss not be less gone 2 seconds and not more than 4 seconds.

3.3.2.4 Terminal velocity. At a distance of 2 inches from battery position, the counter recoil terminal velocity shall not exceed 4 inches/seconds.

3.3.2.5 Rod pull curves. The actual rod curves shall meet the following requirements: see Figure 1)

Range 1: Peak rod pull shall occur within the first 20 milliseconds for long recoil and 60 milliseconds for short recoil.

Range 2: Within 280 milliseconds for long recoil or 180 miliseconds for short recoil, the continuous rod pull curve shall not exceed the peak rod pull of range 1.

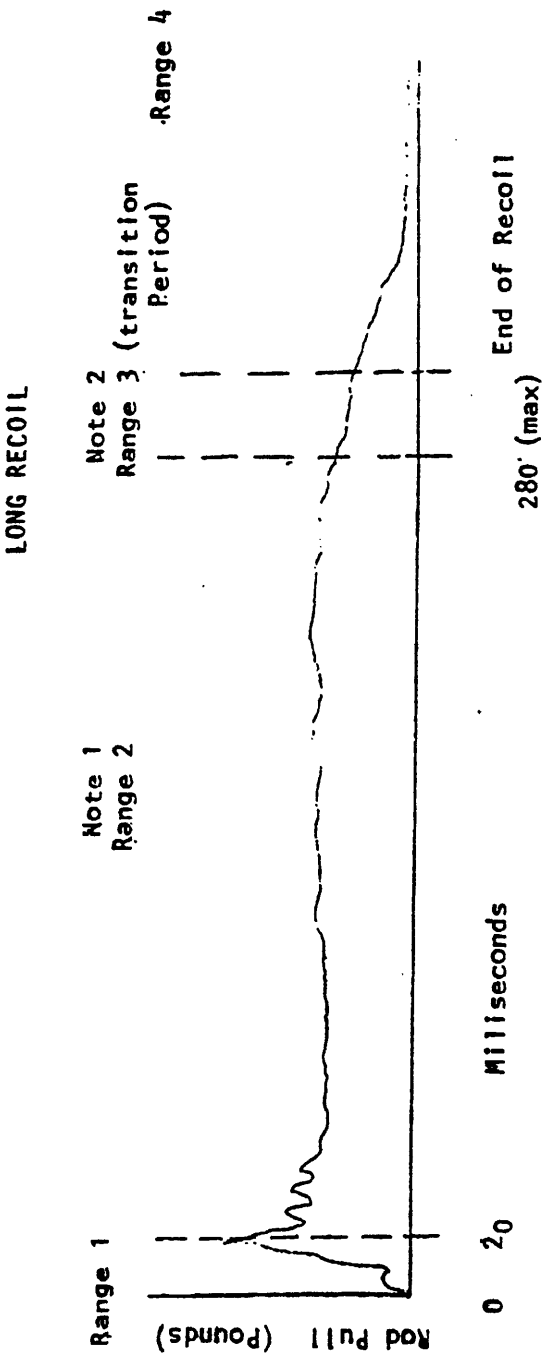
Range 3 After Range 2, the rod pull curve shall be continuoulsy declining to the end of the recoil cycle.

Range 4: Counter recoil cycle. See paragraph 3.3.2.4"

4.6.1 Function firing test: Delete in its entirety and substitute:

"4.6.1 Function firing test. Each recoil mechanism shall be subjected to the powder gymnastication test as specified in Table IV. All testing shall be done on a test stand at zero degrees elevation with the variable recoil locked for long recoil or short recoil, as specified. The tabulated sequence must be followed. Charges shall be varied as required to meet " the stated rod pull requirements and shall impart an impulse of at least 10,000 pounds-seconds to at least one long recoil test mode and the short recoil test mode. If any of the requirements of paragraphs 3.3.2.1 through 3.3.2.5 are not met, the recoil mechanism shall be rejected. Upon completion of this test, the recoil mechanism shall be subjected to the tests specified in paragraphs 4.6.2 through 4.6.7.

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Note 1: Rod pull trace is allowed to increase or decrease within the range as long as it does not exceed the peak in Range 1.

Note 2: Rod pull trace must be continuously decreasing in this range to the end of the recoil cycle.

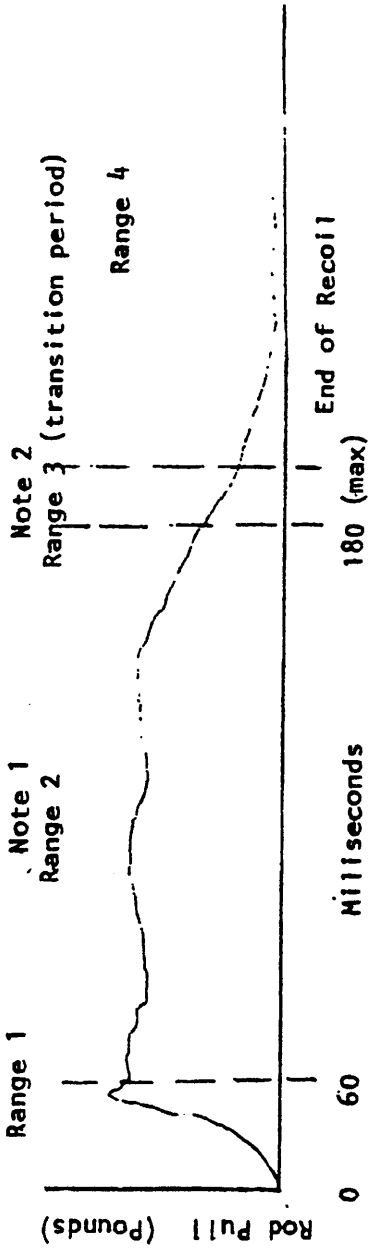


FIGURE 1 TYPICAL ROD PULL CURVES

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Table IV Function Firing

Sequence #	Variable Recoil Setting	Total Required Rod pull (Lbs)
1	Long Recoil	Warmer Max. Rod Pull (Total of both recoil rods) shall not exceed 82,000 pound-force.
2	Long Recoil	
3	Short Recoil	
4	Long Recoil	

PAGE 16

Delete and add new paragraph as follows:

"6.3 One (1) copy of the firing report, which shall include variable data received by this specification and a disitized computer plot of rod pull versus time for at least the first 300 milliseconds of the recoil cycle, shall be sent to AMSMC-QAR-Q (D) and SMCAR-LCW-A, AMCCOM, Dover, NJ 07801 for informational purposes."

Delete original Figure 1 and substitute new Figure 1.

NOTE : The margins of this amendment are marked with an asterrisk to indicate where. changes (additions, modifications, deletions, corrections) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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NOTICE
OF VALIDATION

INCH-POUND

MIL-M-45986(AR)
NOTICE 1
12 October 1990

MILITARY SPECIFICATION

Machine Gun, 7.62 MM M60E2 (Fixed)

MIL-M-45986(AR) dated, 15 February 1977, with Amendment 3, dated 22 October 1984, have been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Army - AR

AMSMC N/A

FSC 1025

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